EXODUS

GUI Controls Guide & Tutorial for v0.1.4

A reference for every button, checkbox, and spin box

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Getting started: A typical session follows this order:
(1) *Load Poni* to set the wavelength,
(2) *Load Files* to import diffraction patterns,
(3) tune background subtraction,
(4) *Load JCPDS* to add crystal phases,
(5) *Fit JCPDS(s)* for a single pattern
(6) *Fit all* for a batch of diffraction patterns,
(7) *Analysis tools* for fitting results.

What's new in v0.1.4

- Pseudo-Voigt peak profile (fixed global η).
- Caglioti peak broadening (refined U, V, W).
- Faster fitting: reflections evaluated only near their centre.
- More robust error estimates;
  Pop-up warning if lattice-parameter errors are undetermined.
- Analysis tools including EoS fit.

1 Data

Load Poni Opens a file dialog to select a `.poni` file produced by pyFAI. The wavelength is extracted from this file and used for all subsequent d-spacing calculations. This button must be clicked before any fitting can begin.

Load Files Opens a file dialog to select one or more diffraction pattern files. The loaded files populate the list on the left; clicking a file in the list – or changing the Pattern No. spin box – displays that pattern in the 1D plots. All current DIOPTAS output formats are supported.

Remove Pattern Removes the selected pattern from the session (the file on disk is not deleted).

Pattern No. (spin box) Scrolls through the loaded dataset by frame number. Changing the value updates the raw and processed 1D plots and keeps the 2D heatmap in sync. A digit-wheel interaction is supported: hover the mouse over a specific digit and use the scroll wheel to increment that digit alone (e.g. scroll on the tens column to step by 10).

Data Range – *from* / *to* (two spin boxes) Set the 2θ minimum and maximum limits applied when loading and fitting data. Any signal outside this window is discarded. After changing these values, click **Recalculate BG** to apply them to the full dataset.

Subtract BG (checkbox) Toggles background subtraction on and off in the processed pattern display without recomputing anything.

Recalculate BG Re-reads all files from disk and recomputes background subtraction using the current 2θ range and BG parameters. Use this after adjusting the data range or BG settings; it does not open the file dialog or clear JCPDS phases or fit results.

2 BG Subtraction

Two mutually exclusive background methods are available:

2.1 ALS Fit

Asymmetric Least Squares (ALS) fits a smooth baseline by iteratively re-weighting residuals to favour points *below* the pattern.

Smoothness ($\times 10^6$) Controls how smooth (i.e. flat) the ALS baseline is. The displayed value is multiplied internally by 10^6. Higher values produce a flatter background; lower values allow the baseline to follow broad curvature. A typical starting range is 1–100.

Width The asymmetry parameter p. Smaller values (e.g. 0.01–0.05) push the baseline closer to the bottom of the peaks; larger values allow it to sit higher. Values must lie in $(0, 1)$.

Iterations Number of ALS refinement cycles. More iterations give a more converged result at the cost of speed; 100 is a sensible default for most patterns.

2.2 Poly Fit

A polynomial background estimated by identifying and masking diffraction peaks, then fitting the residual signal. We recommend this method only for data with well-separated peaks.

Order Polynomial order of the background curve. Increase this to capture more curved baselines.

Prominence Peak-detection sensitivity threshold. Peaks whose prominence exceeds this value are masked before the polynomial is fitted. Decrease to mask weaker features.

Height Minimum absolute intensity for a feature to be treated as a peak. Features below this value are retained in the background estimate.

Width Minimum width (in data points) that a feature must have to be classified as a peak and excluded from the background fit.

3 JCPDS Files

Load JCPDS Opens a file dialog to import one or more `.jcpds` crystal structure files. Each phase appears as a new row in the JCPDS table below. Input allows also for `.cif` files, which are converted to JCPDS automatically; however, the EoS parameters need to be set manually by the user (standard values $K_0 = 20$, $K_0' = 4$)

3.1 JCPDS Table

Each row corresponds to one loaded phase. The columns are as follows:

Column	Description
JCPDS	File name of the loaded phase.
Show	Checkbox – toggles the tick-mark overlay for this phase on the 1D processed plot.
Use	Checkbox – marks the phase as active for Pawley refinement. Only ticked phases are included in fits.
Pressure	Spin box (GPa) – the starting pressure for this phase. The unit cell is updated via the equation of state before each fit.
Start F	First frame number at which this phase is active during batch fitting.
End F	Last frame number at which this phase is active during batch fitting.
Col.	The colour used for this phase's fitted peak markers on the 2D heatmap.
Edit	Opens the JCPDS editor dialog, allowing manual adjustment of lattice parameters, equation-of-state constants (K_0, K_0'), and the reflection list.

Fit JCPDS(s) Runs a refinement on the currently displayed pattern using all phases marked **Use**. The refined fit and tick marks are shown immediately on the processed plot.

Save (single fit) Saves the most recent single-pattern fit to a text file. Columns written are: 2θ, intensity, BG-subtracted intensity, background, fit, and one tick-position column per active phase.

Difference (checkbox) Overlays a difference curve (data minus fit) on the processed plot, useful for judging fit quality.

4 Fitting Parameters

All fitting in EXODUS currently follows the Pawley approach. The minimiser adjusts the unit cell dimensions (a, b, c, α, β, γ) to bring the predicted peak positions into agreement with the data.

The constant background offset C after background subtraction is bounded to $\pm 10\%$ of max $|I|$ by default.

Peak shapes are modelled by default as pure Gaussians, described by a single width parameter σ. This is a deliberate simplification: a full pseudo-Voigt profile requires more parameters per reflection and the evaluation of both $\exp(-x^2)$ and $1/(1+x^2)$ for every point in the pattern for every reflection, versus just $\exp(-x^2)$ for a Gaussian, which is roughly 2x the floating point work per evaluation.

The minimusation of squared residuals are by default set to $ftol/xtol/gtol = 1e{-}5$ and $x_scale =' jac'$. With a least-squares minimiser, the number of Jacobian evaluations scales roughly with the number of free parameters. A rough estimate for a simple case (<10 peaks) is 2–3x faster data processing for pure Gaussians compared to pseudo-Voigts. The Gaussian-only model therefore keeps the parameter count low and the fitting fast and stable, which is the priority for large sequential datasets.

Each reflection is now evaluated only near its centre (± 3 FWHM Gaussian, ± 30 FWHM pseudo-Voigt), which speeds up both profiles.

2θ Min / 2θ Max (spin boxes) The 2θ window passed to the Pawley minimiser. Only data

within this range enters the optimisation, even if the displayed pattern is wider. The window is limited to the 2θ range of the data.

Sigma Initial guess for the peak width σ (degrees 2θ). This value seeds the optimiser; the final value depends on the sigma mode selected below.**Sig. Bounds** is a multiplier on the sigma guess that sets the upper allowed bound during refinement. For example, a value of 2 means σ can refine to at most $2\times$ the initial guess; minimum value is set to 0.

Amplitude A prefactor on the amplitude initial guess, expressed as a fraction of the data maximum. A value of 1.0 starts the amplitude at the tallest point in the pattern; 0.5 starts it at half that height. **Amp. Bounds** is a multiplier on the amplitude initial guess that sets the upper bound. A value of 3.0 means amplitude can refine up to $3\times$ the starting value; minimum value is set to 0.

Max shift % Maximum fractional shift allowed on lattice parameters during refinement (e.g. $0.05 = \pm 5\,\%$). Setting this to exactly 0.0 fixes all lattice parameters, allowing only peak intensities and widths to refine.

4.1 Peak Profile (mutually exclusive checkboxes)

Gaussian Pure Gaussian peaks (default).

Pseudo Voigt Mixing of Lorentzian and Gaussian with one FWHM; global η determines the mixing (0 = pure Gaussian, 1 = pure Lorentzian); η is fixed at the GUI value for speed reasons.

4.2 Peak Broadening (mutually exclusive checkboxes)

Fix Sigma Refines one global peak width for all peaks throughout the fit. Useful when peaks are poorly resolved or the dataset is noisy.

Fix sigma $1/\cos(\theta)$ (default) Applies a $1/\cos\theta$ angular broadening dependence. A single width parameter is refined globally and the per-peak width is computed from it. This is the recommended starting point.

Vary Sigma Refines an independent peak width for every peak. More flexible but requires good signal-to-noise.

Vary by phase Refines one peak width per phase, shared across all peaks of that phase. A good compromise when different phases have noticeably different intrinsic broadening.

Caglioti Peak width for any reflection follows the equation $\mathrm{FWHM}^2 = U\tan^2\theta + V\tan\theta + W$ with global U, V, W. The values are refined from the start values in the spin boxes.

5 2D Plot and Batch Fitting

5.1 Batch Fit Modes (mutually exclusive checkboxes)

Sequential Fitting Each frame is fitted starting from the refined unit cell of the previous frame. The structural evolution is propagated automatically through the dataset – ideal for continuous compression or heating runs.

Pressure Walk Each frame's starting unit cell is predicted by incrementing the pressure from the previous frame's fitted value by **Delta P**. The pressure step between frames used in Pressure Walk mode.

Pressure Guess Uses a manually entered frame-vs-pressure table (one tab per phase, populated with the + button) to interpolate starting pressures for each frame. Suitable for non-monotonic or multi-phase experiments where pressure points are known from ruby fluorescence or similar.

5.2 Batch Action Buttons

Fit all Launches the batch fit using whichever mode is selected. A progress dialog appears with a **Cancel** button. Each frame is processed in turn and results accumulate in the 2D overlay as fitting proceeds.

Save All Saves all sequential fit results to a chosen directory: one CSV of lattice parameters and pressures per phase, plus one text file per frame containing the full pattern, background, fit, and per-phase tick positions.

Save 2D Plot Saves the 2D heatmap – including any fitted peak overlays – as a TIFF or PNG file.

Lin / Log / SQRT Switch the 2D heatmap's intensity colour scale between linear, logarithmic, and square-root normalisation. Log and SQRT scales help reveal weak features alongside strong peaks.

Colour Cycles through available colormaps for the 2D heatmap.

5.3 Analysis tools for fitting results

As of version 0.1.4, the following additional features were implemented. However, we would like to point out that these are currently only in their beta-test stage, and - if used - should be tested thoroughly against well established programs.

Plot Data Plots the fitted results (e.g. P, V per frame) with errors.

EoS Fit (beta) Fits an isothermal EoS (BM2, BM3, BM4, Vinet, natural strain, Murnaghan) to the P–V data; shows refined / fixed / implied parameters.

Stress/Strain Placeholder – not implemented yet.